

2022 Ph H2 Q16

A cyclist and bicycle of total mass 72 kg travel up a hill at a constant speed.

(a)
Calculate the component of the weight acting down the slope

Use:
 $W_{\text{parallel}} = mg \sin \theta$

- Given:
- $m = 72 \text{ kg}$
 - $g = 9.8 \text{ m/s}^2$
 - $\sin \theta = \sin(7.5^\circ) = 0.1305$

$$W_{\text{parallel}} = 72 \times 9.8 \times \sin(7.5^\circ) = 72 \times 9.8 \times 0.1305 = 91.9 \text{ N}$$

(b)
Calculate the total force acting against the motion of the cyclist

The forces acting against motion are:

- Frictional force = 28 N (given)
- Component of weight down the slope = 91.9 N (from part (a))

$$F_{\text{total}} = 28 + 91.9 = 120 \text{ N}$$

(c)
Calculate the useful power output of the cyclist

Use:
 $P = Fv$

Where:

- $F = 120 \text{ N}$ (total opposing force, since motion is constant speed)
- $v = 4.2 \text{ m/s}$

$$P = 120 \times 4.2 = 504 \text{ W}$$

 Revision Tips

- Weight component **down a slope**: $mg \sin \theta$
- Constant speed $\Rightarrow$ total driving force = total opposing forces
- Power = force $\times$ speed when force acts in the direction of motion
- Be careful to **add friction and weight components** when total resisting force is needed